

PROGRAMMING QUESTION PAPER
Round 3 - Application Development
Time Allowed: 2 Hours 30 Minutes

Question:

Write a Java console application for an online shopping cart system called Z-Kart. Z-Kart is an online shopping platform focusing on selling electronic gadgets such as laptops, mobiles, and tablets. The application should have the following functionalities:

1. Customer Initialization
2. Inventory Initialization
3. New Account Signup
4. Password Encryption
5. Login
6. Shopping
7. Admin Mode
8. Attractive and Unique Deals for Every Customer
9. Deal of the Moment
10. Persistence of Order History
11. Force Password Change
12. Password Complexity and History

1. Customer Initialization (zusers_db.txt)

The backbone for every online retail application is its customers. The initial customer details will be given in a file called zusers_db.txt. Each line in the file is of the following format:

Format: Email EncryptedPwd Name Mobile

Email	Password (Encrypted)	Name	Mobile
abc@zoho.com	ApipNbjm	Rahul	9922992299
123@zoho.com	Cboljoh	Anitha	8564119904
user@zoho.com	kbwb22	Arpan	9872345693

2. Inventory Initialization (z-kart_db.txt)

The inventory of Z-Kart is initialized from a file called z-kart_db.txt. Each line in the file is of the following format:

Format: Category Brand Model Price Stock

Category	Brand	Model	Price (Rs.)	Stock
Mobile	Apple	6S	60,000	10
Mobile	Motorola	G	12,000	5
Laptop	HP	Elite	56,000	20
Tablet	Google	ChromeBook	8,000	12

3. New Account Signup

There must be a provision to sign up new accounts. While adding new customers, collect their email id, name, mobile number, and password. Ask the user to re-type the password to avoid typos and ensure both passwords match before proceeding. Since every customer must be identified uniquely, ensure the given email id does not already exist in the Z-Kart database. If the email already exists, prompt the user to sign in instead. Append the newly added customer details into the `zusers_db.txt` file.

4. Password Encryption

It is not advised to store passwords directly in files. The encryption rule is simple - add +1 to each character of the plain password. The encryption supports wrap around as shown below. Always encrypt the password before writing it to external storage.

Plain Char	Encrypted Char	Rule
a - y	b - z	lowercase a to y -> +1
z	a	lowercase z -> wraps to a
A - Y	B - Z	uppercase A to Y -> +1
Z	A	uppercase Z -> wraps to A
1 - 8	2 - 9	digits 1 to 8 -> +1
9	0	digit 9 -> wraps to 0
0	1	digit 0 -> wraps to 1

5. Login

The program starts with a Login Page. The user enters their email and password. The program encrypts the typed password using the +1 rule and compares it with the stored encrypted password in `zusers_db.txt`. If both match, the user is allowed to proceed to shopping. If the email is not found, the user is directed to sign up.

6. Shopping

After login, the user selects a product category such as Mobile, Laptop, or Tablet. The system displays all products available under the selected category. The user can select products and specify the quantity. They can continue shopping across multiple categories before proceeding to checkout. The invoice shows all selected products, individual prices, and the total amount. Before completing the purchase, the system checks if sufficient stock is available. If stock is available, the order is placed and the stock is decreased in z-kart_db.txt. If stock is not available, the system shows an Out of Stock message.

7. Admin Mode

The Admin is a special user with extra access to manage the system. The Admin can view and update product stock quantities in z-kart_db.txt. The Admin can view and manage all registered user details in zusers_db.txt. The Admin can force a password change for any user and process password reset requests.

8. Attractive and Unique Deals for Every Customer

Every customer must be shown attractive and unique deals when they log in. These deals are personalized based on the customer profile and purchase history. The system must generate and display relevant offers to encourage the customer to buy more products.

9. Deal of the Moment

Apart from personalized deals, the system must also show a Deal of the Moment. This is a special time-based offer that is available for a limited time only. The deal is shown to all customers and is updated periodically.

10. Persistence of Order History

Every successful order placed by a customer must be saved permanently. The order details including customer email, products purchased, quantities, prices, and total amount must be written to an order history file. This data must persist across sessions so that past orders can be viewed at any time.

11. Force Password Change

The Admin has the ability to force a password change for any user in the system. When a user's password is flagged for forced change, the user must change their password on their next login before they are allowed to access the shopping features. The new password must meet the password complexity rules.

12. Password Complexity and History

When a user sets a new password, it must meet the following complexity rules. The password must contain uppercase letters, lowercase letters, digits, and special characters. Additionally,

the system must maintain a password history for each user. The user must not be allowed to reuse any previously used passwords.

ALL THE BEST !!!